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Acknowledgements.

Stick some more acknowledgments here.

Abstract

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Contents

1 Introduction **1**

1.1 Importance and Applications of Soft Robotics 1

1.2 Types of Self-Healing Polymers 3

1.3 Comparison of Different Self-Healing Pathways 5

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway 6

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway 7

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway 7

1.4 Applications of Self-Healing Polymers in Soft Robotics 8

1.4.1 State of the Art Self-Healing Soft Robots 8

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics 9

1.5 Limitations with Current Materials for Soft Robotics 9

2 Project Aims and Objectives **10**

3 Synthesis of a Self-Healing Polymer **11**

3.1 Results 11

3.1.1 Hydrolysis of the Polyimide 11

3.1.2 Spectrophotometry 14

3.1.3 Cross-linking of the Polyimide 15

3.1.4 Catalytic Imidisation of polyimide 18

3.2 Experimental 19

3.2.1 Equipment 19

3.2.2 The Hydrolysis of a Polyimide 20

3.2.3 Spectrophotometry as a Determination Method for Hydrolysis . . 20

3.2.4 Crosslinking of Polyamic acid 21

3.2.5 Catalytic Imidisation of Polyamic Acid 23

4 Construction of a Bellows Based Actuator 24

4.1 Design 24

4.1.1 Construction of a Folded Protective Sleeve 24

4.1.2 Complete Assembly 29

4.2 Results 30

4.2.1 Stiffness Characterisation 30

4.2.2 Temperature Stability of the Bellows 32

4.2.3 Workspace Characterisation 34

4.2.4 Navigation 35

4.2.5 Clinical Case Study: Drug Delivery in Stomach 35

4.3 Methodology 37

4.3.1 Stiffness Characterisation 37

4.3.2 Temperature Stability of the Polyimide Sleeve 37

4.3.3 Workspace Characterisation 38

4.3.4 Moulding of polyamic acid 38

4.3.5 Moulding of Polyamic Acid 38

List of Figures

1.1 A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[1] 2

1.2 a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer 4

1.3 a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene. 6

4.1 caption 30

4.2 A) The Kapton sleeve is placed into an oven for an hour and then placed onto the stiffness characterisation experimental setup. B) The initial profile of the actuator before heating. C) The profile after 3 cycles in the oven. D) A plot comparing the average force vs. displacement across all cycles 33

4.3 caption 36

List of Tables

3.1 ATR-FTIR absorbance ratios and relative imide surface content for various samples in comparison to Wu (2017)[2] 13

3.2 ATR-FTIR peak positions, absorbances, and relative imide surface content 19

3.3 Dye mixtures for examination in a spectrophotometer with dilution by number of parts water. Here a "part" is defined as 1.5mL. 20

3.4 Samples for testing in a spectrophotometer using dye mixture summarised in Table 3.3. 21

3.5 Catalysts, reaction temperatures and analytical methods used in crosslinking attempts of polyamic acid with ethylene carbonate. 22

Chapter 1

Introduction

1.1 Importance and Applications of Soft Robotics

Soft robots are a type of robot known for their use of soft materials, compliant nature and near infinite degrees of freedom . Degrees of freedom (DOF) is a term related to the number of independent movements a robot can make in 3D space.[3] Soft robots are usually nature inspired such as "vine-like" tip growing robots,[4,5] "snake like" continuum robots[6,7] and "hand-like" soft grippers.[8,9] The advantages of using a body that is highly deformable and with near-infinite DOF means these devices can navigate tortuous environments with ease.

This report focuses on continuum robots, often characterised by their long flexible bodies, commonly used for navigation and delivery or retrieval of objects in constrained environments.[10] Continuum robots can be classified into "continuous backbone" and "segmented back-bone" designs.[10] Continuous backbone devices are made from a soft and compliant material, in this report we will call these "soft-bodied", whereas, a segmented backbone refers to those that are made from hard, inflexible segments that are flexibly attached to achieve high curvatures, referred to here as "hard-bodied".

Hard-bodied continuum robots benefit from stiffer materials allowing a greater ease of

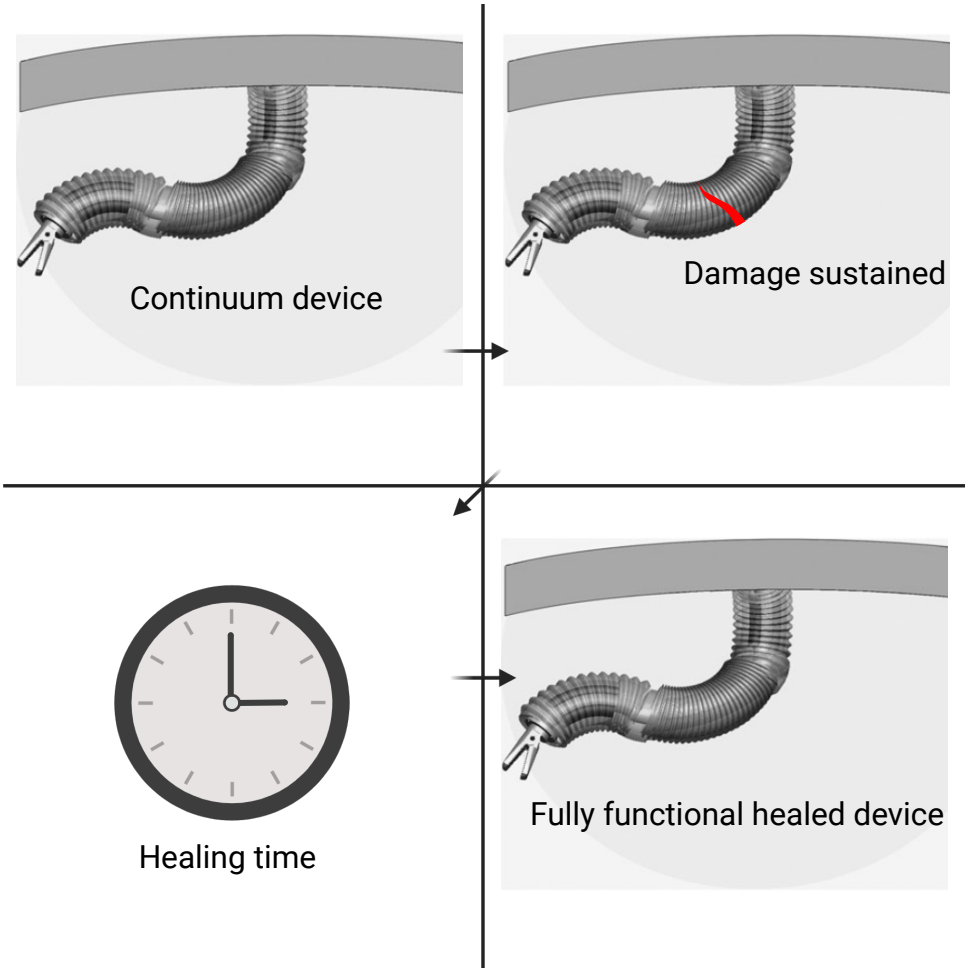


Figure 1.1: A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[1]

control[10] which can improve dexterity. However, they cannot deform and are more likely to impart a greater force on the surrounding environment.[11] Hard continuum robots are often tendon driven, whereby the segments are joined by wires that can be pulled to give specific movements. Whereas soft bodied continuum robots can achieve high curvatures and low forces on surroundings,[12] but are often harder to control due to their methods of actuation[10] and can suffer from material failure.[13] Soft continuum robots commonly take the form of an inflated body, these can be controlled in many ways such as varying pressure,[4] strain limiting a specific region[14,15] or magnetic actuation.[5,11] Pressurised systems that rely on fluids need a constant seal to maintain function as opposed to a segmented tendon driven approach that can be partially damaged and still actuatable. This leads the development of resilient materials such as self-healing polymers that can be applied to soft robotics.

1.2 Types of Self-Healing Polymers

Polymers are a type of molecule that consists of several repeating sub-units, these form chains that are combined many times over to form materials. Self-healing (S-H) polymers are a type of polymer that can heal damage.[16] This occurs through reactions and interactions between multiple polymer chains to reconnect chains that have been shifted apart through damage. The process of damaging or cutting a polymer does not break bonds in the polymer backbone but instead can break interchain interactions called cross-linking, allowing the chains to drift apart, reforming the cross-linking interactions will "heal" the material.[17]

S-H polymers can be classified into intrinsic or extrinsic healing, and autonomous or non-autonomous.[18] Materials with an intrinsic healing capacity involve a healing mechanism through chemical bonds or interactions that are continuously present in the polymer, rather than extrinsic healing, such as capsules of "healing" material that are broken through damage and causing the material to reform.[19] Autonomy refers to the healing

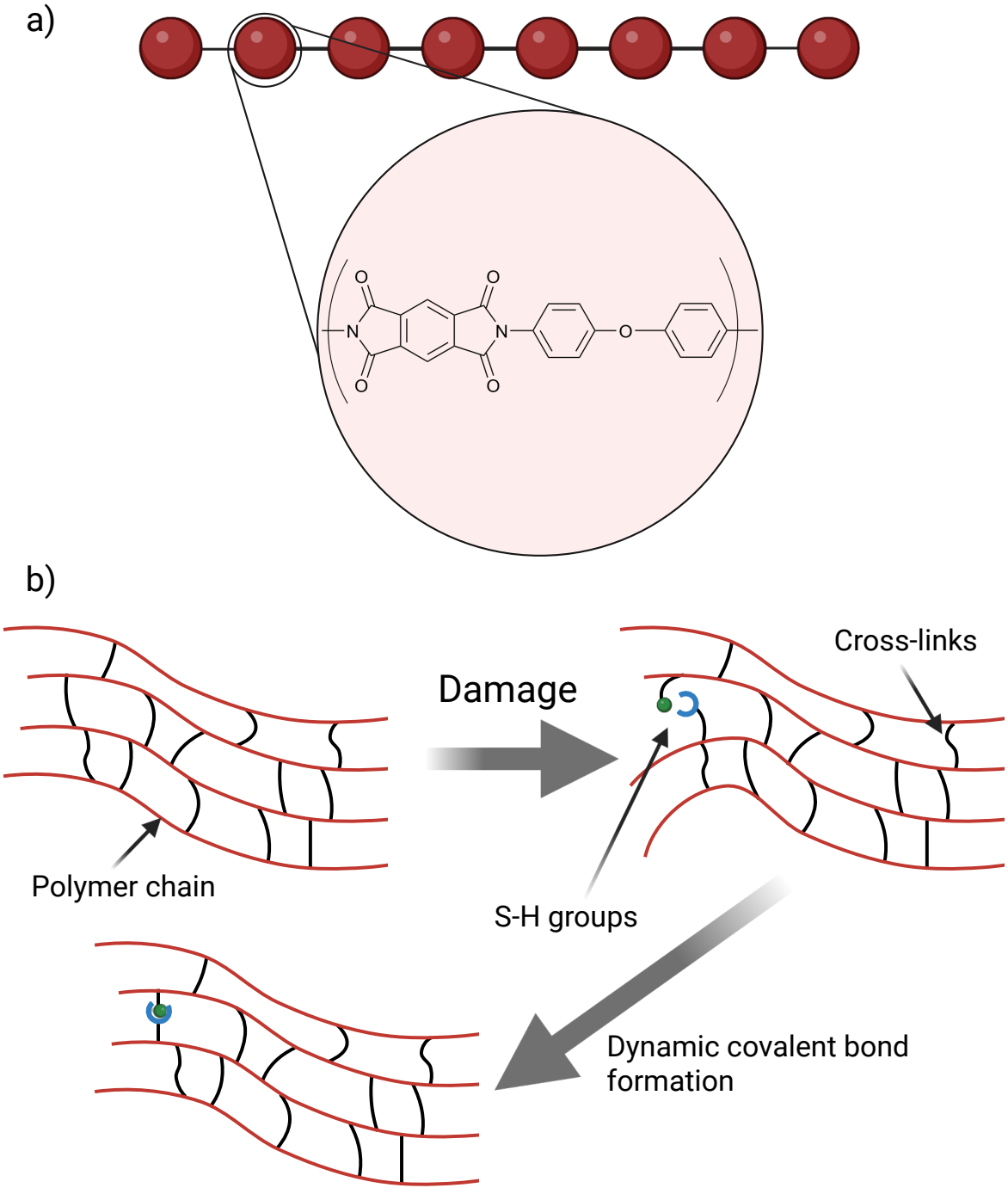


Figure 1.2: a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer

process occurring without external input. For example, capsules that burst upon damage of the material and release a healing agent are classified as autonomous, whereas a photochemical reaction that needs UV light to activate[20,21] is a non-autonomous mechanism.

1.3 Comparison of Different Self-Healing Pathways

There are a variety of different pathways that can be taken to create a S-H polymer depending on the chemistry of the initial polymer. The pathways that will be focused on are intrinsically self-healed types, however, both autonomous and non-autonomous types will be considered.

The three pathways below rely on a property of some electrons called conjugation. Conjugation occurs when specific electron orbitals overlap in such as way that allows the electrons to move around the molecule. This is often exploited with a property of some atoms called electronegativity, having a highly electronegative atom in a molecule will attract conjugated electrons to create a local high density region of electrons and therefore a polarisation in a molecule.

Dynamic covalent bonds can form and break based on the conditions applied to them[22] and they are ideal for use as mechanisms for S-H due to their intrinsic nature and reliability in production of high yields.[23] This report investigates the use of two types of dynamic covalent bonding, Diels-Alder and Michael additions, and an example of a supramolecular interaction. Supramolecular chemistry is based on interactions that are non-covalent and may be more suited to a soft material.[24] This is due to the S-H mechanism of the cross-links that forms interactions therefore the polymer chains can move semi-independently.[24,25] Whereas, a dynamic covalent bond has a much higher bond energy[22] and can therefore get closer to the original physical properties of a polymer.

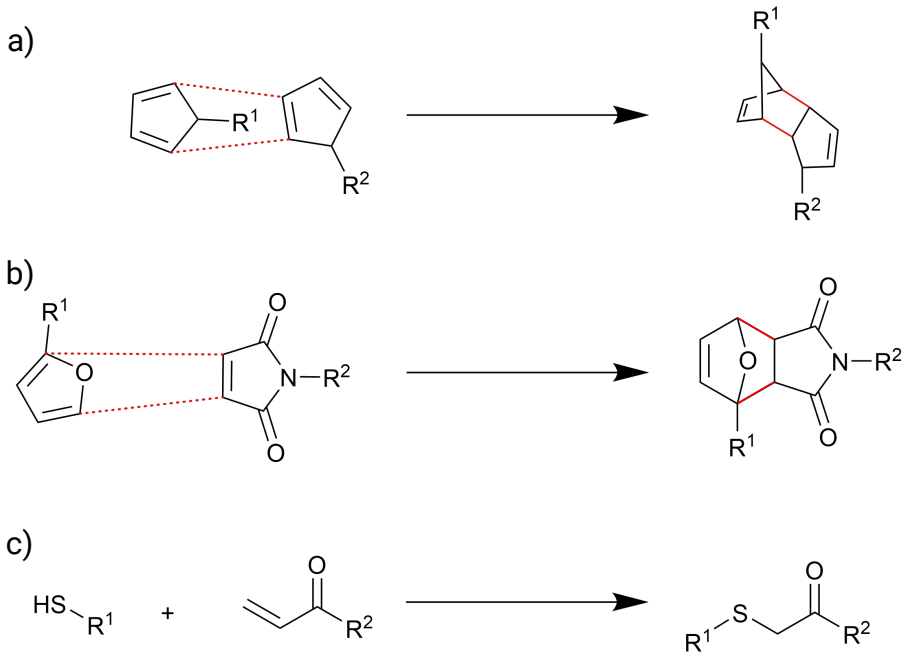


Figure 1.3: a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene.

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway

Diels-Alder (DA) reactions require a diene, a class of molecule characterised by having two sets of double bonds, and an alkene ("dieneophile").[22,26] The reaction occurs when conjugated electrons from the diene are free to move to form new bonds with the carbons either side of the alkene double bond, this creates a new ring structure. A faster reaction is generally observed with an electron rich diene and an electron poor dienophile.[27] The electronic character of the reagents can be altered by changing their local substituents. This class of reactions can have lots of different conditions depending on the molecules involved. Some variations can occur at room temperature,[28] which may make them an ideal choice for an autonomously healed medical device as the healing can occur whilst deployed in the body. However, many DA reactions are light[20,21] or heat activated.[29]

Some molecules such as cyclopentadiene can perform a DA reaction with itself, this has the potential to be an ideal choice for a S-H polymer as the molecule will never be in close proximity to a molecule that it cannot react with. However, some evidence has shown

that the bonds formed are not very strong and therefore high tensile strength materials may not be possible.[26]

One of the most common types of DA reaction is a heat activated reaction between a derivative of both furan and maleimide.[30] They react extremely accurately and require mild conditions,[26] therefore they are very suitable for application as a robotics material. One of the reasons that this reaction is so effective is that the maleimide has electron withdrawing carbonyl groups which make the dieneophile electron poor and therefore reacting faster. However, this mechanism has a disadvantage compared to cyclopentadiene as there are two halves of the end product. Therefore, there must be a two different processes occurring to create both the furan substituted and maleimide substituted chains. This increases the number of total steps in the reaction which increases the risk of low reaction yields and wasted material.

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway

Thiol Michael additions are a type of reaction that relies on a thiol, a molecule characterised by a sulfur containing group and a conjugated alkene.[31] The conjugated alkene used in a Michael addition is with a carbonyl group that will attract the electrons from the alkene to allow the thiol group to react. This has a similar advantage to the Diels-Alder reaction that it is not totally dependant on specific reactants merely groups of reactants. Therefore slight changes can be made to the adjoining groups to tweak the reaction conditions. This has been previously shown to have been successful with a heat activated reaction that could heal scratches after 15 to 30 minutes at 60 °C.[32]

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway

Pi-Stacking is a type of intermolecular interaction whereby conjugated systems interact based on the neighbouring substituents determining if a region is electron rich or poor.

These electron disparities cause an electrostatic interaction. This method commonly uses pyrenyl groups, these are polycyclic aromatic compounds, a very pi-electron rich species. These can be interacted with a napthalene diimide complex, an electron poor pi-system to give four pi-stacking interactions. A method has shown the creation of a polymer chain end group which can sandwich a napthalene group within two pyrenyl groups.[33] This method can be combined with other intermolecular interactions like hydrogen bonding[34] or mettalosupramolecular interactions to give a stronger interaction.[35]

1.4 Applications of Self-Healing Polymers in Soft Robotics

Continuing the theme of biological inspiration of soft robotics, S-H polymers can be used to extend the lifespan of components. In soft continuum robots, the goal is usually to make the robot softer than the environment surrounding it.[12] This presents a challenge that the environment can often be abrasive and can damage the device. It is therefore evident that a damage resilient material would be useful. Damage resilient materials can take multiple forms: Damage isolation, material elasticity, physical self healing such as low melting temperature materials and chemical self healing.[36]

1.4.1 State of the Art Self-Healing Soft Robots

State-of-the-art approaches using intrinsic S-H mechanisms are commonly based on DA chemistry.[18] These have been successfully implemented in tendon-driven[37] and pneumatically controlled soft robotic devices.[38,39] Such devices are capable of recovering from macroscopic damage, including punctures and cuts. Additionally, there are non-covalent approaches that utilize hydrogen bonding in the development of soft self-healing actuators, such as in McKibben muscle designs.[40]

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics

Terryn *et al.*[18] states that there are 5 main criteria for a S-H polymer; macroscopic damage must be healed completely; healing can take place multiple times at the same location; there are minimal material property changes after healing; elastomeric behaviour is achieved; the reprocessing and recycling of a material is possible. Here we propose that elastomeric behaviours are non-essential and a more useful requirement would involve a timely S-H reaction.

1.5 Limitations with Current Materials for Soft Robotics

Damage control whilst a device is active can be a challenge,[41] this shows a need for autonomous S-H.[18] Non-autonomous S-H can be made "functionally autonomous" through the use of stimuli responsive mechanisms, however, the current options are slow in operation.[41] Currently recovery timescales of applied S-H polymers are around 25 hours.[36,39] By altering the polymer used or S-H pathway this could be improved. There is also a lack of an easy synthesis route for many of the current polymer options,[36] thereby increasing the cost and reducing widespread use. Also the number of damage-repair cycles in these devices are low, such as 3 or less.[39] Most devices do not achieve all 5 criteria laid out in the Terryn *et al.* review,[18] such as the McKibben design of A. Lopez Diaz *et al.*[40] only reports a healing efficiency of 65%.

Chapter 2

Project Aims and Objectives

Chapter 3

Synthesis of a Self-Healing Polymer

3.1 Results

3.1.1 Hydrolysis of the Polyimide

The method used was based on previous literature publications[2,42–48] where an alkaline solution is used to hydrolyse polyimide to form a salt intermediate and then an acidification to form carboxlyic acid groups. A NaOH solution (0.2M)[42–44] was initially trialled for the alkaline hydrolysis step at room temperature for 90 mins, however when there was minimal change under Infrared spectrometry (IR), the temperature of the reaction was increased to 120°C. Also, the NaOH concentration was adapted similar to that used in Xue (1986).[47] This comprised of the use of 1 equiv. OH^- for every 1 monomer of the polyimide. This was chosen to ensure a partial hydrolysis, to only expose one carboxylic acid group rather than the total hydrolysis to form a tetrasubstituted benzene ring. However during analysis of this reactio we also observed minimal hydrolysis via IR.

A widely accepted base-catalysed amide hydrolysis route is through a second order pathway, where a hydroxy group (OH^-), from the base, nucleophillically adds to the carbonyl group (C=O) to form a tetrahedral intermediate.[49] This cleaves into two and the amine

is eliminated. In pthalamide structures, a molecule close in character to the polyimide monomer, has been shown to exhibit similar behaviour when in solution.[50]

However, mechanistic changes occur when the reaction contains macromolecules, it has been shown in previous literature that there is a third order dependence on the concentration of OH^-([OH^-]) and follows a mechanism that starts with the nucleophillic addition of the OH^- to form a charged intermediate before undergoing rate-determining cleavage of the C-N bond to form the associated polyamic acid.[42] Moens (1957) proposed that, due the hydrostatic repulsion, the rate of hydrolysis is lowered when a polymeric chain has increasing numbers of negative charges,[51] this would imply that the formation of the monanion intermediate could hinder the reaction.

While multiple literature sources opt to use KOH,[2,45,47,48] hydrolysis has been shown to occur in the presence of NaOH.[42–44] In retrospect, hydrolysis attempts using KOH should have also been approached due to the success of many KOH hydrolysis attempts.

However, we attribute most of the failure in this experiment to the low concentration of NaOH, as sources have stated that a concentration below 0.5M was ineffective.[2] This explains why the testing with 0.04M was also ineffective. Additionally, some of the failure may be caused by the lack of addition of a reagent to maintain constant ionic strength, which has been shown to assist hydrolysis with NaOH.[42]

Spectral Analysis

Structural determination was conducted via Infrared spectroscopy (IR). A comparison was made between the attempted hydrolysis of the sheets and a purchased form (polyamic acid).

Characteristic peaks of both the hydrolysed and the control forms are observed around 1500cm^{-1} , corresponding to the C=C bond in the central benzene ring[2,45] along with the 1235 cm^{-1} aromatic ether stretch[45] both which do not change throughout the hydrolysis.

Sample	Base	Conc. /M	A_{1500}	A_{1720}	A_{1720}/A_{1500}	Rel. imide content
90 min, 25°C	NaOH	0.2	0.2243	0.2304	1.0273	0.9424
20 min, 120°C	NaOH	0.03	0.1062	0.1167	1.0990	1.0082
40 min, 120°C	NaOH	0.03	0.1608	0.1735	1.0789	0.9898
Kapton	—	—	0.1125	0.1226	1.0900	1.0000
PAA	—	—	0.8508	0.1735	0.2040	0.1871
Wu (2017)	KOH	0	0.0513	0.0587	1.1440	1.0000
Wu (2017)	KOH	0.1	0.0504	0.0573	1.1375	0.9943
Wu (2017)	KOH	0.5	0.0511	0.0568	1.1096	0.9699
Wu (2017)	KOH	1	0.0587	0.0617	1.0519	0.9194
Wu (2017)	KOH	5	0.0763	0.0072	0.0946	0.0827

Table 3.1: ATR-FTIR absorbance ratios and relative imide surface content for various samples in comparison to Wu (2017)[2]

Also observed is the C=O asymmetric stretching mode at around 1710 cm^{-1},[2,43,45–47] the peak around 1775 cm^{-1} as the symmetric C=O stretch[45–47] and the C-N-C imide ring stretch at 1370 cm^{-1},[45,46] all of which decrease in magnitude during hydrolysis.[2,46]

One literature source[52] states that the symmetric C=O stretch is at 1775 cm^{-1} and the asymmetric is at 1720 cm^{-1} however, due to the disparity between this and the other sources, we disregarded this result.

Some sources estimate their degree of hydrolysis with a comparison of the ratio of intensities of the C=O : C=C (1720:1500 cm^{-1}) peaks,[2,47] these are summarised in the table 3.1. In this report, the symmetric C=O peak was observed slightly lower at around 1710 cm^{-1}.

As in the Wu 2017 paper,[2] this report assumes the Kapton sample to have 100% imide content on the surface, hence finding that the polyamic acid sample contains 18.71% surface imide content. Also observed was the decrease in imide content relative to both an increase in NaOH concentration and time. This work found that the best hydrolysis attempt was the 90 minute, 0.2M hydrolysis at room temperature, giving a total imide surface content of 94.24%, meaning that 5.76% of the imide rings were cleaved.

3.1.2 Spectrophotometry

Due to the limitations of only using one method of analysis of the degree of hydrolysis of Kapton, an additional method was sought out. Other traditional methods of determination such as nuclear magnetic resonance (NMR) rely on solubilty of samples, this is a property which, by design, Kapton does not have. Therefore, a spectrophotometric test was proposed.

Other spectrophotometric methods of detection of carboxylic acids involve fluorescent tagging. These involve the attachment of a fluorescent label to the carboxylic acid group, producing a distinct peak in a UV-vis spectrum. Attachment reagents involved are varied, including halogenoalkyl, hydrazine, amine, alcohols and sulfonate groups. However, most of these options require activation of the COOH group before binding of the fluorescent marker. We aimed to avoid any pre-activation reagents as this would lower the activation of the unwanted intramolecular amide formation, thereby reforming an imide ring.[53]

Therefore, a method proposed by Johar (1972) involving the use of Eosin Y (EY) and Methylene blue (MB) for determination of carboxylic acids was used.[54]

Dyes were mixed as according to the concentrations stated by Johar (1972). However, is was apparent that ther concentration was too high, therefore additional dilutions of the dye mixtures were made. Initally tested were the interactions between the dyes and water, we observed a peak at (515 ish) confirmed to be corresponding to EY[55–57] and a peak at (650 ish) correspoding to MB.[58–61] The shoulder on the EY peak is formed through the dimerisation of the molecule due to the high concentration.[56]

An example of a postive test would occur with acetic acid, a simple carboxylic acid. In the spectophotometer, the peak corresponding to the EY was destroyed. This report proposes that this destruction is due to the use of glacial (the most concentrated an acid can be) acetic acid. In high concentrations, acetic acid can to convert the EY into a precursor (dihydrofluoran)[62] which causes the observed blue shift in the peak[63] along

with the reduction in magnitude.

Next assessed was the interaction between dye mixture and polyamic acid (PAA) dissolved in dimethyl sulfoxide (DMSO). This gave two peaks, one in the EY region at (515ish) and one in the MB region at (650ish). The interaction with the MB was very similar to that of the acetic acid which was a promising result. However, in the EY region, the large peak was not similar to that of acetic acid. We propose this effect was caused by EY catalysing amide formation, hence forming imide rings and removing the carboxylic acids. This implies that the peak at (515ish) corresponds to the solvent (DMSO) rather than an interaction with the PAA.

To test this hypothesis, DMSO was mixed with the dyes and a peak at (515ish) and (650ish) was observed, which we confirm to be EY[56] (maybe aslo cite aroteinvestigating) and MB[58,60](maybe also cite tchaikovskaya2023solvent). As a final validation, we checked that the interaction was not simply pH based by testing HCl, this was chosen as an acid that does not contain carboxyl groups and therefore should react differently to the acetic acid. The absorbance spectrum showed one peak at (450), which we believe is due to the ....

Due to the difference in absorbance for the acetic acid and the PAA and the catalytic effects of EY on imide reformation, we determined that the GS Johar method is not appropriate for a determination of hydrolysis of Kapton.

3.1.3 Cross-linking of the Polyimide

Cross-linking attempts follow a route where ethylene carbonate (EC) is used to form an ester with the carboxylic acids on neighbouring polymer chains, thereby joining them together with a "cross-link". This was attempted with three different salts (Et_4NBr,$ZnCl_2$, $FeCl_3$) as catalysts.

During the cross-linking reaction with Et_4NBr, an orange-brown suspension was formed

that when washed, filtered and freeze dried yielded a pale orange powder. With $ZnCl_2$, the liquid was a dark orange clear liquid that, when water was added, gave a single clump of grey polymeric residue. This was freeze dried to remove the excess water and then analysed. The cross-linking reaction with the $FeCl_3$, yielded a dark orange/brown gel, which formed a grey gum-like texture in small pieces when water was added. This was also frozen and analysed.

During a cross-linking with ethylene carbonate, CO_2 is released. This was a concern as the monowave reactor has a pressure limit of 20 bar. To ensure there was a reliable safety buffer, an artificial limit of 5 bar was set. The Ideal Gas law was used to give the amount of EC to use to ensure this limit was held.

$$PV = nRT \rightarrow n = \frac{PV}{RT}$$

P = 500 kPa, V = 10 mL, R = 8.314 J K^{-}1 mol^{-}1, T = 423K

$$n = \frac{500 \times 10 \times 10^{-3}}{8.314 \times 423} = 0.00142mol$$

Since the CO_2 is generated from the EC, there are same number of moles for the CO_2 and for the EC. This motivates the use of $mass = Mr \times n$, where Mr = 88, n = 0.00142

$$Mr \times n = 88 \times 0.00142 = 0.12496g$$

Detailed mechanistic studies of the reaction between a carboxylic acid and ethylene carbonate in the presence of a quaternary ammonium salt, such as Et_4NBr, have been performed in other literature.[64–66] Yoshino (1977) states that the mechanism begins with a nucleophillic ativation by the anion of the catalytic salt on a methylene group of the EC, this cleaves the hydrogen from the alcohol group (OH) of the carboxylic acid which can

now nucleophillically substitute the halide to form the ester. Fagerburg (1984) disagreed with this mechanism as they proposed that the rate of the reaction was be dependent on the anion of the catalytic salt. However they did not find this experimentally. They observed during a reaction of the catalytic salt and EC alone, the activation energy of the EC decomposition was much higher than any esterification attempts . They therefore proposed a mechanism involving the attack of a carboxylate species on the methylene carbon of the EC to decompose the ring and form the ester product. Given the depth of Fagerburg's work, we consider this mechanism to be more robust than the one proposed by Yoshino.

The change of catalyst to a lewis acid ($ZnCl_2, FeCl_3$) was to encourage a direct activation of the carbonyl group on the EC, the mechanism for this proposed step was as follows: The lewis acid activates the carbonyl on the EC by forming a "complex". This allows the alkoxide to attack the methylene group of the EC and form the hydroxyethyl ester.[67] Von Seggern (2020) however reported, the use of zinc based catalysts was found to be ineffective in this reaction.[67]

Spectral analysis

Spectral analysis was performed mainly with IR, but also with ^{1}H NMR and DOSY.

From the IR data, the crosslinking with Et_4NBr was deemed to be unsuccessful as the product spectra the key peaks of the C-O bond corresponding to an ester or any alcohol formation were not observed. Alcohol peaks would be likely as crosslinking attempts are unlikely to be 100% successful, hence the single ester product would also be formed. However, neither of these were present and since the product was insoluble, no NMR could be obtained. This IR spectrum was compared with the spectrum of Kapton (i.e. the ring closed form of the polyamic acid) and the similarities confirmed that the imide ring reformation had occured. Key spectral peaks confirming the lack of crosslinking contain the C-N-C peak, commonly found at 1370 cm^{-1}, however in our sample we observed this

peak at 1391 cm^{-1}. Other signs of imidisation include the increase in the magnitude of the C=O stretch at 1715 cm^{-1}. These peaks would indicate the imide ring reformation as opposed to the esterification product.

From the IR data of the $ZnCl_2$ and the $FeCl_3$ attempts, we observe two new peaks at around 1800 cm^{-1} which are not in either the Kapton or the PAA samples. This shows the presence of EC in the final sample as C=O bonds in liquid EC[68] are found in this regon. Since these samples were soluble, proton nuclear magnetic resonance (^{1}H NMR) was available. A new peak was observed at 4.43ppm in the ^{1}H NMR spectrum, however, when adding 2 μL of EC in a "spike test", the peak increased in size confirming that the new peak was starting material (EC). Other peaks in the $ZnCl_2$ and $FeCl_3$ samples are indicative of unreacted polyamic acid. After increasing the temperature of the $ZnCl_2$ catalysed sample, the peaks positions show the intramolecular cyclisation of the imide ring with the C-N-C stretch at 1370cm^{-1}.

DOSY data was collected for the $ZnCl_2$ sample ....

3.1.4 Catalytic Imidisation of polyimide

During the attempted crosslinking reactions, the imide ring formation was observed at a lower temperature than previously seen in the literature. This motivated attempts to recreate this with only polyamic acid (PAA) and $ZnCl_2$.

Without the catalyst, material got slightly darker after heating for an hour. This was attributed to some minor solvent evaporation and therefore, substrate becoming more concentrated. A similar colouration after heating the sample containing catalyst was observed, however, there was also material suspended. This was proposed as indicative of a partially successful attempt.

An assessment of the success of the imidisation can be made by comparing the A_{1720} to the A_{1500} in the same manner as previous analysis.[2] In table 3.2, again the Kapton sheet was

Sample	ν_{1500} /cm^{-1}	A_{1500}	ν_{1720} /cm^{-1}	A_{1720}	A_{1720}/A_{1500}	Rel. imide content
PAA	1495.02	0.8508	1709.18	0.1735	0.2040	0.1871
No_cat	1495.02	0.4209	1717.34	0.1375	0.3268	0.2998
With_cat	1497.06	0.0727	1721.42	0.0443	0.6084	0.5581
Kapton	1497.06	0.1125	1709.18	0.1226	1.0900	1.0000

Table 3.2: ATR-FTIR peak positions, absorbances, and relative imide surface content

assumed to have 100% surface imide content and others were calculated relative to this. This report finds that the imide content of the polyamic acid heated without the catalyst had 29.98% imide content as compared to the baseline value from the unheated PAA at 18.71%. However, the sample with $ZnCl_2$ catalyst presents as successful as 55.81% imide content is observed. Also observed was a shift in peak position in the 1500 peak when imide content is over 50%. To confirm the degree of imidisation in the future, a pure sample could be obtained through a recrystallisation from DMSO.

3.2 Experimental

3.2.1 Equipment

A MonoWave reactor (MonoWave 50 Closed Reaction Vessel; ANTON PAAR) was used for the majority of experiments covered in this report. This reaction vessel contains integrated temperature and pressure control and a magnetic stir element. This allowed the user to perform reactions more quickly as samples could be heated above their ambient pressure boiling points. The MonoWave reactor required a specific 10mL tube that was sealed with a rubber cap and a PTFE septum, to ensure agitation, an oval-shaped magnetic stir bar was included.

Dye Mix	Parts water	Parts Eosin Y (0.5%)	Parts Methylene Blue (0.1%)
Mix 1	10	2	1
Mix 2	5	2	1
Mix 3	0	2	1

Table 3.3: Dye mixtures for examination in a spectrophotometer with dilution by number of parts water. Here a "part" is defined as 1.5mL.

3.2.2 The Hydrolysis of a Polyimide

Polyoxydiphenylene mellitimide (Kapton) film (0.18g, thickness: x) was cut into pieces of approx. $3mm^2$ and placed into a beaker. This was stirred at room temperature with a solution of NaOH in DMSO (0.2M, 22ml). There was no observable difference and an Infrared (IR) spectrum was taken.

Another sample of Kapton film was cut into approx. $3mm^2$ pieces and placed into a MonoWave reaction tube along with a solution of NaOH (vol, 0.03M). This tube was placed into a MonoWave reactor that was set to ramp for 20 minutes and hold for 20 minutes at 100°C. There was no observable difference, so a sample was taken and the remaining mixture was placed onto the MonoWave under the same conditions. The mixture was filtered and thoroughly washed with water before IR was collected.

3.2.3 Spectrophotometry as a Determination Method for Hydrolysis

The preparation of samples for the spectrophotometric analysis was as follows:

3 mL 0.5% Eosin Y solution was added to 1.5mL 0.1% Methylene Blue solution. This solution was diluted with 10 parts water and 5 parts water (where a "part" is 1.5mL) to give three dye mixes (summarised in 3.3)

Test compound (2.7mL) was added to a sample vial along with one of the dye mixes (0.3 mL) to form samples summarised in 3.4. These were inverted to ensure mixing before

Sample number	Test compound	Dye mix
S01	Dimethyl Sulfoxide	Dye 1
S02	Polyamic Acid solution	Dye 1
S03	Water	Dye 1
S04	Acetic Acid	Dye 1
S05	Hydrochloric Acid	Dye 1
S06	Dimethyl Sulfoxide	Dye 2
S07	Polyamic Acid solution	Dye 2
S08	Water	Dye 2
S09	Acetic Acid	Dye 2
S10	Hydrochloric Acid	Dye 2
S11	Dimethyl Sulfoxide	Dye 3
S12	Polyamic Acid Solution	Dye 3
S13	Water	Dye 3
S14	Acetic Acid	Dye 3
S15	Hydrochloric Acid	Dye 3

Table 3.4: Samples for testing in a spectrophotometer using dye mixture summarised in Table 3.3.

being transferred to a plastic cuvette and placed in a spectrophotometer (DETAILS) to test between 250 and 750nm.

3.2.4 Crosslinking of Polyamic acid

This procedure outlines the crosslinking procedure of polyamic acid (PAA) via ethylene carbonate (EC) using a catalytic salt as outlined in previously published procedures.

A mixture of polyamic acid (0.51mL, conc) was placed into a MonoWave tube with ethylene carbonate (0.08g) and a catalytic salt. These were placed into the MonoWave to ramp for 20 minutes and hold at the reaction temperature for an hour. The products were washed with water and filtered before being freeze dried and examined on the spectrometers. The reagent amounts and processing methods are summarised in table 3.5.

Catalytic Salt	Mass Salt /g	Reaction Temp. /°C	Analytical Method
Et_4NBr	0.08	150	IR
$FeCl_3$	0.02	100	IR, NMR
$ZnCl_2$	0.03	100	IR, NMR, DOSY

Table 3.5: Catalysts, reaction temperatures and analytical methods used in crosslinking attempts of polyamic acid with ethylene carbonate.

Tetraethylammonium Bromide (Et_4NBr) Crosslinking Spectral Data

A orange powder (grams,moles,yield) was isolated, ν max/cm^{-1}(solid); 1715, 1601, 1501, 1391, 1307, 1252, 1167, 1116, 885, 815, 800, 724 634 606, 567, 516, 428

Iron Chloride ($FeCl_3$) Crosslinking Spectral Data

A white gum (grams,moles,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid); 2931, 1805, 1774, 1721, 1609, 1497, 1403, 1372, 1309, 1256, 1163, 1114, 1075, 987, 822, 759, 724, 679, 606, 563, 479

Zinc Chloride ($ZnCl_2$) Crosslinking at 100°C Spectral Data

A grey gum (g,mol,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid); 3263, 3112, 1679, 1603, 1583, 1530, 1493, 1403, 1367, 1316, 1263, 1187, 1163, 1081, 1036, 1008, 836, 722, 706, 630, 600, 559, 534, 502, 404

Zinc Chloride ($ZnCl_2$) Crosslinking at 110°C Spectral Data

Brittle brown pieces (g,mol,yield) were isolated, ν max/cm^{-1}(solid); 3043, 1776, 1715, 1606, 1538, 1497, 1371, 1307, 1238, 1165, 1114, 1012, 879, 820, 724, 632, 606, 565, 514, 430

3.2.5 Catalytic Imidisation of Polyamic Acid

Polyamic acid (0.51mL), $ZnCl_2$ (0.02g) and a magnetic stirbar were added to a MonoWave tube. This was placed in the MonoWave at 110°C to ramp for 20 minutes and hold at 150° for 60 minutes. A control experiment without the $ZnCl_2$ was also placed under the same conditions. The control experiment gave a clear dark brown viscous liquid which, when dried, gave a gum-like pale orange solid. The test with the $ZnCl_2$ gave a similar dark brown liquid, however this was slightly cloudy and, when processed, gave an orange powder.

Control Spectral Data

IR(solid): vals, vals cm^{-1}

Catalytic Imidisation Spectral Data

IR(solid): vals, vals cm^{-1}

Chapter 4

Construction of a Bellows Based Actuator

4.1 Design

Actuators were constructed using a combination of a folded polyimide sleeve and a internal multilumen silicon tube with nitinol wires for actuation. Our methodology is inspired by similar "bellows-style" approaches to construct continuum devices (references) with the key difference in the use of Polyimide sheets (Kapton) forming the main body of the device.

This work focuses on the use of Kapton (thickness =) due to its good robustness and high temperature capabilities.

4.1.1 Construction of a Folded Protective Sleeve

Protective sleeves are a common addition to soft actuators due to the damage prone nature of soft materials and also to increase the functionality for use in harsh environments such as high heat, flames, harsh chemical environments and space. Nanomaterial based

coatings are often at the forefront resistance with examples based on flame and heat resistance and others for chemical resistance through harnessing superomniphobia[69] (resilient to almost all liquids). However nanomaterials are only suitable as coatings rather than the main building material, therefore the use of resilient polymers improves this. In this report, we use Kapton to form the main body of the protective sleeve. Kapton has been used before in protective sleeves and has been shown to survive harsh environments such as space. Most famously, the surface of Eagle, the Apollo 11 lunar landing module, was coated in Kapton film due to its thermal insulation properties. In this report, we construct a Kapton sleeve to be used for a resilient soft robotic device.

Folded bellows style actuators have been demonstrated as One of the harshest environments that robotics are commonly deployed in is space, materials for space robotics require lightweight, high compressibility, extreme temperatures, high UV radiation.[70] In the body, chemical resistance can be particularly useful with pHs of stomach acid being as low as 1. We opt for Kapton film as the main building material of our actuator due to its chemical, UV and heat resistance allowing a contructed device to be applicable to a wide number of scenarios.

In order to achieve resilient structures, the chosen material needed to be available in film material, flexible, temperature resistant, chemical resistant Kapton and other polyimides contain ring structures that allow the partial break down of the polymer chain for the addition of intrinsic functionality without the complete conversion back to oligomer material.

There three main designs of folded cylinders used in soft robotics: Kresling, Yoshimura and Accordion. Possible the most popular design is Kresling, which was discovered through the twist buckling of paper cyclinders,[71] this was adapted to form consitent folds through the segmentation of a flat sheet into triangles to form a helical folding pattern.[72] Miuri patterns take the form of.... Yoshimura patterns are formed through the segmentation of a sheet into parallelograms in a diamond configuration, these parallelograms are bisected with horizontal lines to form triangles.[73] Yoshimura designs can

be modified by the inversion of specific protruding vertices to form a pattern known as Accordion. This transforms the parallelograms into hexagons and therefore the triangles into trapeziums. It has been previosuly shown that for an effective "wrinkle-free" compression, the triangles in Yoshimura designs should be right angles[74] and hence, the hexagons in Accordion should have a horizontal short side length equal to $\sqrt{2}a$ and long side length to be $2\sqrt{2}a$ if the angled sides are of length a.

An in depth analysis of Accordion versus Kresling was performed by Butler (2016),[70] in which they found that the number of folds to make a functional device was approximately $10\times$ less for the Kresling design. Often the initial failure of these devices occurs on the vertices, which is also lower in the Kresling. Also, when comparing compressibility (using Eq. 4.1), they find that while both designs have a a compressibility over 90%, the Kresling design slightly outperforms the Accordion by 1.5%.

$$compressibility = \frac{h_{deployed} - h_{compressed}}{h_{deployed}} \tag{4.1}$$

Due to the configuration of the nitinol wires in the silicon actuation device, we found that the controllability was improved when the device had torsional stability. While there are methods of adapting a Kresling design to limit torsional movement through the reflection of alternating layers. The devices are inherently prone to torques through the helical design. Accordion Yoshimura designs are significantly more torque resistant in comparison. This is one of the primary reasons for our use of the Accordion folding pattern.

Origami structures have been shown to minimise the buckling of catheters.[75]

The actuator exhibits self-deploying characteristics, which are present in other devices. This is a product of treating the material as having non-zero thickness.[76] Steps can be taken to avoiod thios by adjusting the location of the folds. however when flding by hand, there is so much precision that moving the folds by the thickness of the width of a sheet

is not neccessary (there is a reference somewhere)

To manufacture an Accordion patterned sleeve, a 2D drawing to map the fold lines was drawn in Fusion360 and printed onto paper. This paper was taped to the table and a sheet of Kapton was taped over the top, the Kapton is clear so the printed lines are visible through. The lines were then traced onto the Kapton with a ruler and a ballpoint pen. The ballpoint pen marked the desired fold lines and also partially indented on the lines. The sheet of kapton was removed from the table and the outer edge was cut with scissors. The Kapton sheet was then folded along the lines as shown in the figure [ref figure]. The folds on the sheet were puished back out and the kapton was rolled into a cylinder. This was taped along the overlapped edge to form a temporary seal while maintaining care to line up the folds. The tape was removed in small sections and JBweld HighHeat was applied until the whole adhesion region. This was threaded onto a plastic tube with a diameter approx. 2mm smaller than the kapton sleeve. The sleeve was taped to the tube and the adhesion region was taped over the top to clamp the glued region. This was left for 24 hours to cure. The glueing and the curing process was performed in a Fume cupboard [brand/details] due to the risk of respiratory irritation through the curing process. After curing, the sleeve was removed from the plastic tube and a strip of Kapton tape was placed over the edge on the adhesion region. This improved the flexiblity of the region. one of the ends was folded over to make a cap and this was also glued. In this work, this assembly is known as the "Kapton sleeve" or "polyimide sleeve".

Manual folding versus using a plotter

A commercially available plotter (Cricut®) was used with a scribe tool and a cutter to achieve a rapid manufacturing of a folded sleeve. This was attempted with paper initally for cheap prototyping and then with the polymer film. This would avoid the time consuming folding steps

The Cricut was set up with the cricut design space software. This software allows the

user to draw and communicate with the cricut either through bluetooth or via cable. The cricut has an adhesive cutting mat to hold the material in place whilst the tool interacts with it. For our testing, we drew a rectangle (10 x 5 cm) and set the cricut to cut on this line. Then a line (5cm) was drawn to split the long edge into two squares (5cm^2), this line was set to score. These two drawings were attached so as they would be performed in the same material loading cycle, the scoring was set to be first in the cricut's "job queue" to avoid there being potential for the score tool to catch the newly cut edge of the polymer and causee the adhesion between the sheet and the polymer to fail. A piece of the desired material was adhered to the cutting mat and the cricut was started. This method was repeated for 2 thicknesses of Kapton and 80gsm paper (roughly 0.25mm).

The three pieces of cut material were taped by the top edge to the jaws of a vice. The vice had the jaws enlarged by 3D printed pieces to ensure that the clamping would occur across the whole of the material. The vice was manually controlled to compress the pieces of material into the crease to form a fold and then slowly decompress to evaluate the fold. The folding with the paper was highly effective, when the jaws were opened the paper was clearly separated into two sections by a fold in the centre. Both thicknesses of Kapton were not so effective, when they were decompressed there was no clear fold line and the material was left haning in a curve. The plotter can visible score the Kapton, however, the score lines do not define a crease as the scoring tool does with paper. This phenomenon is explored further in the [put paper here], where polyimide is shown to not form precise folds.

We determine here that the use of a plotter could be useful in the rapid production of folded sleeves, however in its current form, the cricut is not a viable route for creasing Kapton. A comprimise could be to use the cricut to draw the fold lines by using a pen in place of the scribe tool, this would allow the user to bypass the printing and tracing the pattern by hand.

4.1.2 Complete Assembly

The complete assembly was formed with the polyimide sleeve, the silicon continuum device and 3D printed parts.

The silicon continuum device contains a silicon 7-hole multilumen tube with a 3D printed cap, the lumen are arranged so that there are 6 equally spaced edge lumen and one central lumen. There are 3 nitinol wires that are approximately double the length of the silicon tube and these are threaded through one of the edge lumen, through the 3D printed cap and back through a neighbouring lumen. This is repeated another 2 times until all 6 outer lumen contain a niinol wire. We define the disatl end as the end with the 3D printed cap. At the proximal end, where the ends of the nitinol wires are, these are fed into 3 3D printed handles where the wires are held with 1.5mm grub screws. This assembly is threaded through a stiffer tube (approx. 15mm diameter) and a cable gland on the end of this stiff tube holds the silicon tube and the larger tube together. The device can be actuater by fixing the stiff tube in place and pulling on the 3D printed handles. In some instance, we increased the stiffness of the device to improve the controllability of the deivce by putting a nitinol wire through the centre lumen. This centre lumen can also be used for adding to the functionality of the device by presenting an opportunity for targeted drug delivery, a tool channel or for wires from tip-based components.

The construction method for the completed assembly is as follows: the silicon continuum device is threaded through the middle. The glued end of the kapton sleeve has a cable gland (size and details) and a 3D printed octagonal connecting piece glued to it. the silicon device is threaded through the middle of the sleeve and secured in place with the cable gland. The other end of the sleeve is clamped to the silicon continuum device with 3D printed parts to form the completed device.

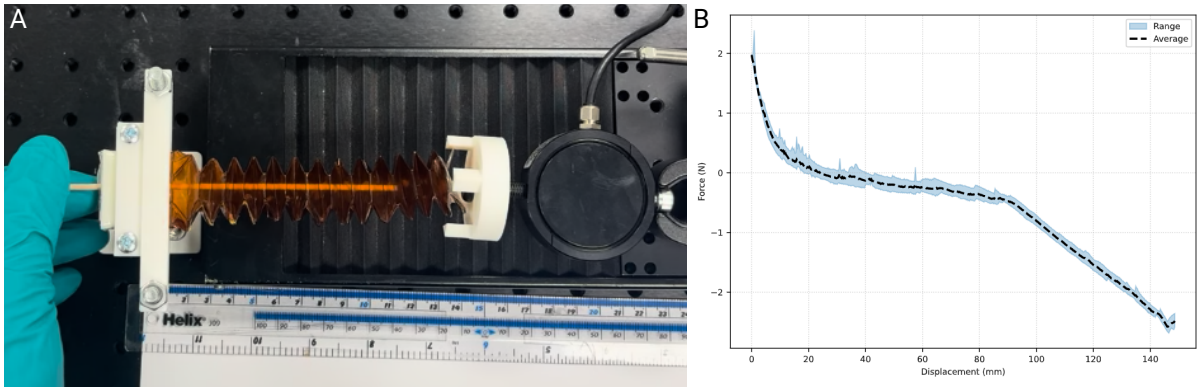


Figure 4.1: caption

4.2 Results

4.2.1 Stiffness Characterisation

The compression of the device was observed to be modelled by an exponential plot, this gave a peak force of x relative to zeroing at a relaxed length of 120mm. The kapton sleeve is acting as a non-linear spring. Origami structures have been shown to act as nonlinear springs[77] when a uniaxial force is applied. Reid (2017)[76] test the force under uniaxial loading for an accordion style device constructed from Acetate (thickness:0.1mm). Instead of a manually folded construction, hinges are created by perforating the material with a laser cutter.

Due to the design process being time consuming, a limited number of kapton sleeves were contructed. Combined with the fact that the sleeves were fragile and prone to failing on the corners, the device was not consistently air tight. This meant that the unpressurised stiffness was characterised, this is an uncommon method and due to the number of parameters related to the design of the sleeve, a direct literature comparison was not available.

Also, the linear stage should have been set to go out and back because we have assumed that the force would be the same but it might not be, also it would do something interesting with the zero point which shifts.

Because the compression of the device causes the folds to settle, the zero point when expanding occurs earlier than expecting meaning the relaxed length changes. If the linear stage was set to go in, then out and then back in, there would be a full profile of extension and compression

The constraint method of a rod up the inside caused the slight wiggles in the plots, although care was taken to stay away from the load cell attachment, the experimental set-up could have been varied to include bars that would slip inside the mount avoiding the need for the manual constraint.

However, the non-linear force profile of this device implies that when the device is extended most it applies the least linear force. This means that during actuation, the device will want to return to its relaxed length. Hence, the pressure required to extend when the sleeve is constrained to a length shorter than 120mm will be lower than if it was acting as a linear spring. This unpressurised characterisation sets a low bound on the stiffness of the device when pressurised the device will exert more force axially. The implication of the settling behaviour of the folds is that, in the future, the developmetn of a control platform would have to take this into account.

Seen in figure 4.1

The stiffness of the polyimide sleeve applies an additional force on the silicon actuation device. For instance, when the device is pressurised, the additional forces applied to the device from the stiffness of the sleeve will affect the amount of air pressure needed to vary the actuators length.

As the force from compression and extension of the actuator was obtained on one experimental set up, the force in extension is a negative force.

The extension of the device was limited

This was only obtained for 60mm past the relaxed length as extension further than this made folds on the actuator invert and reversible retraction was not possible.

The load cell was zeroed at 90mm displacement, however when the actuator reached 90mm in length the force measured was -0.5N. We believe this is due to the folding in the sleeve having some elastic characteristic [cite the kapton folding paper] and therefore taking up the folds and making the relaxed length temporarily shorter.

The maximal force under compression was (x) and the tensile force was (y). during compression it appears that an exponential would fit and during extension the line is linear.

Other devices have often used the Kresling design, Polyimide sleeves using the kresling design show values of x

Due to the unique nature of this device, direct literature comparison for a similar sleeve with similar material and similar testing material is unavailable. This is due to the fact that most other sources have successfully made an airtight bellows which can be tested for the change in pressure versus force.

4.2.2 Temperature Stability of the Bellows

To evaluate the stability of the acutator, tis was placed in the oven. After removing and running the sample 3 times after an hour each time. We determined that the actuator was very temperatrue stable.

Data from the load cell was processed in python and plotted in graphs. The plots were trimmed to start once the force was over 0.2N. This gave plots as shown in 4.2.

This report found that there was minimal change to the performance of the kapton sleeve after heat treatment, it was observed that the variability on results is less than x when calculated using python This is due to kapton having a thermal decomposition at x temperature, hence the lack of behavioural change after heating to 200°C is unsurprising. Kapton based structures have been shown to resist many thermal cycles before degrading[70] Limitations of this experiment are similar to the stiffness characterisation tests due

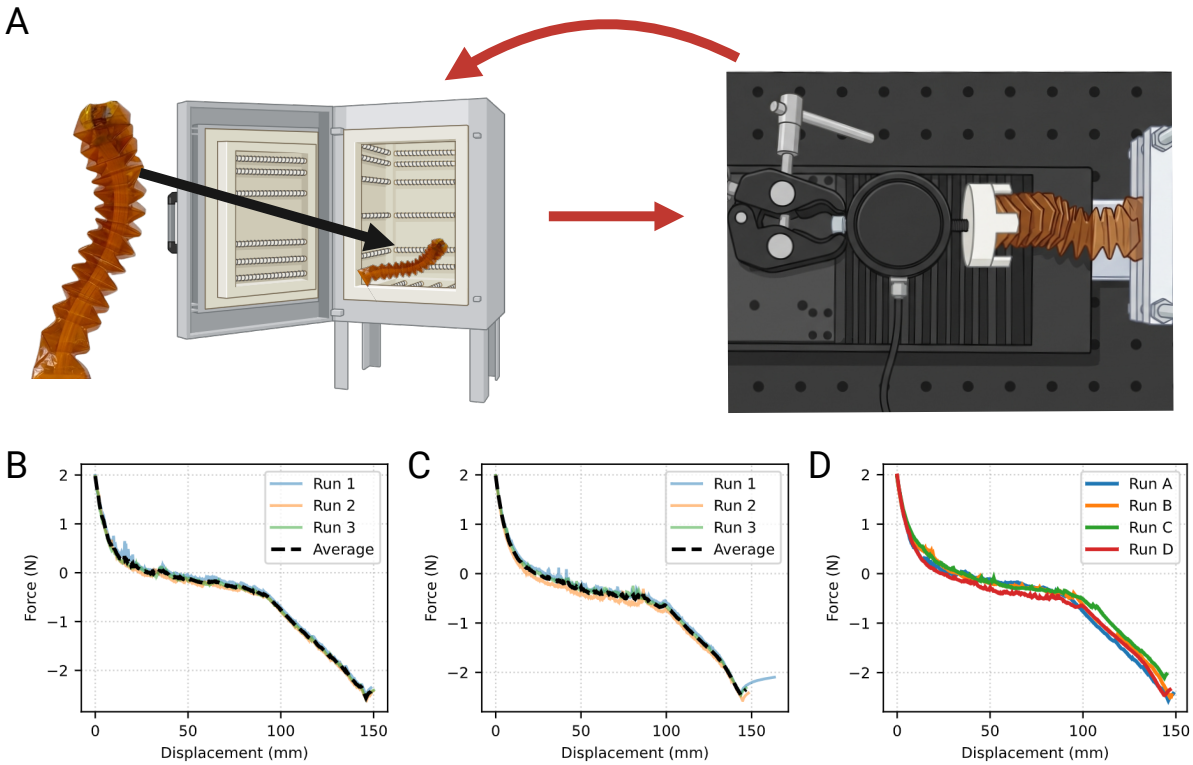


Figure 4.2: A) The Kapton sleeve is placed into an oven for an hour and then placed onto the stiffness characterisation experimental setup. B) The initial profile of the actuator before heating. C) The profile after 3 cycles in the oven. D) A plot comparing the average force vs. displacement across all cycles

to the use of the same experimental set-up

The lack of behavioural change implies that the assembly would be well suited to a thermally harsh environment. Paired with other properties such as the resistance to UV degradation this motivates the use of polyimide as a material for space robotics.

4.2.3 Workspace Characterisation

Workspace characterisation of a device is important to gain an understanding of the total regions that a device can get to

The workspace of a device is the set of all reachable tip positions that the device can achieve.[78]

We tested 4 different assemblies which gradually increase the stiffness to demonstrate the stabilisation of the polyimide sleeve to aid the controllability of the silicon actuation device.

The workspace of the device was found as a hemisphere with a radius proportional to the length of the actuator. When increasing the stiffness of the device with nitinol and the kapton sleeve, the variation in paths was smaller

When adjusting lenght, the device showed a rotation in the z axis. This was due to the method of length adjustment. Since adjustment is done by clamping with a cable gland, the twisting to tighten also rotated the silicon actuator and therefore when pulling the handles in the marked order, the rotation of the device caused the actuation to perform differently.

The device could not achieve a nice circle The folded sleeve was shown to improve the controllability of the silicon actuator by increasing the torsional stiffness.

4.2.4 Navigation

To demonstrate the robots ability to navigate an arbitrary path, a route was set up using 3D printed obstacles to simulate a path that a real world use case could need. The goal of the path was to show that fluid could be delivered through a channel in the silicon inner tube.

This path started shorted than the relaxed length of the actuator and so required the operator to pull on the silicon tube to retract the device before steering the device using the tendons.

Extending the device further than the relaxed length was difficult due to the soft the silicon tube buckling under compression. This meant that the maximum path length was limited to around 120mm.

Practically the effects of the stiffness of the bellows are felt, when we use our device for naviagation we found that retracting the silicon and extending is by pushing it was easy until the relaxed point. After this, overcoming the force of the sleeve pushing back on the silicon actuation device causes buckling internally due to the lack of rigidity of the silicon tube.

The introduction of a pressurised approach to controlling the locomotion of the device is beyond the scope of this

The device successfully navigated the pre-planned path taking x seconds to complete the x cm journey with a success rate of 100%.

4.2.5 Clinical Case Study: Drug Delivery in Stomach

Finally, we demonstrate the use of the bellows based actuator in a stomach training simulator for the application of targeted drug delivery. Here we use the same assembly as the previous navigations with the addition of water to simulate the drug delivery. We use

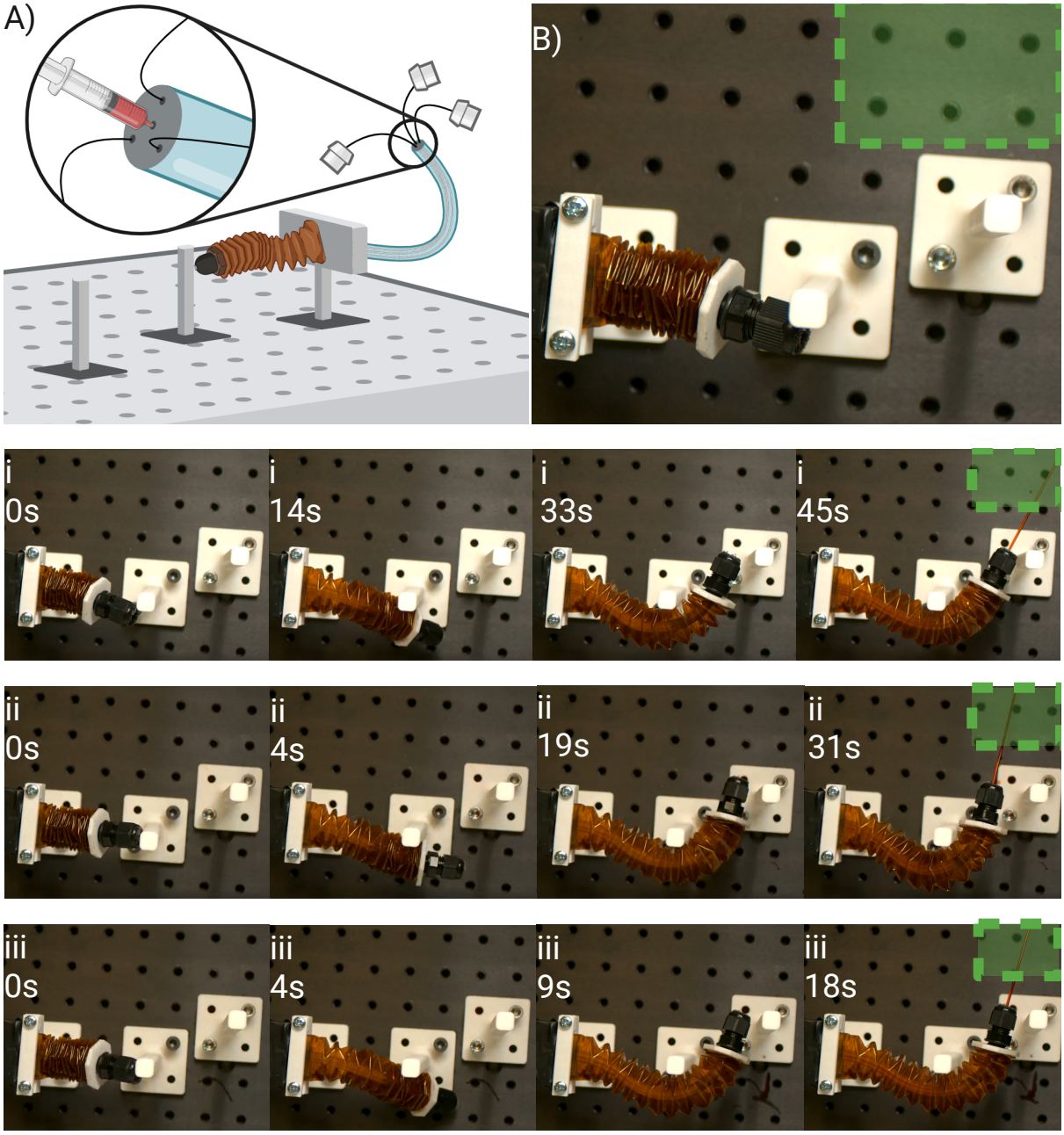


Figure 4.3: caption

an external camera (camera details) inserted through a separate entry into the stomach to give an approximate location for completing the navigations.

4.3 Methodology

4.3.1 Stiffness Characterisation

To characterise the stiffness profile, a linear stage was used with a load cell (brand) held in a SmallRig Magic Arm. This was attached to the end of the polyimide sleeve which was clamped in a 3D printed stand. The whole experimental setup was attached to a ThorLabs Breadboard. The linear stage was set to 60mm (the resting length of the actuator was 120mm, the linear stage is in reverse so to get to 60mm, the linear displacement is 90) and the load cell was zeroed. The load cell was moved to 150mm (actuator length is 20mm compressed) and then the linear stage was set to go to 0mm at 0.5mm s^{-1}. This contained a compression force and a stretching force. The speed of the actuator was translated into the displacement using:

$$v = \frac{d}{t}$$

. The load cell data was collected using DSCUSB Toolkit and processed in python.

4.3.2 Temperature Stability of the Polyimide Sleeve

Using the same experimental set-up as the stiffness characterisation. The kapton® sleeve was placed in an oven at 200°C for 1 hour, left to cool and then compressed and stretched as before.

4.3.3 Workspace Characterisation

To characterise the usable workspace of the device, an experiment was set-up with the four camera Optitrack system and Motive 2.3 software to track the tip position of the device in 3D space with submillimeter accuracy (mean error: value). When assembled, the device was not very predictably controllable partially due to the additional weight at the tip of the added cable gland. This meant that the actuator

4.3.4 Moulding of polyamic acid

Polyamic acid undergoes an imidisation reaction at 150-200° to form Kapton. It shoudl, in theory be prossible to pour the polyamic acid into a mould and heat to form Kapton. This was attempted by pouring the PAA into a mould and heating to 150°C for 20 mins to cause imidisation. This was repeated after placing half the mould into the smartvap with a 3D printed adaptor. The smartvap heats the sample and blows a stream of air into a vortex over a sample to remove the solvent. This was performed for a week before then heating the sample to 150°C for 1 hour.

4.3.5 Moulding of Polyamic Acid

Polyamic acid in NMP (stats) was poured into a Cone-shaped mould. This was assembled with a 3D printed part to connect to the SmartVap (Supplier) for 1 week at 100°C. This reduced the volume and made the liquid a darker brown. This was heated to 150°C for 1 hour to cause imidisation before being extracted from the mould with a scalpel. This experiment was also repeated without drying off the solvent first. Undergoes thermal imidisation

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